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[AI-02] - AI Audit Report

Student Information

Student name :	NGÔ NGUYỄN THÊ KHOA
Student ID:	23127065
Class / Cohort:	23KTPM3

Assignment ID:	HW02
Assignment date:	02/06/2026 - 29/06/2026
AI tool(s) used:	GitHub Copilot, Antigravity

Audit-01

Anchor ID: #audit-01

1 Prompt + Tool

GitHub Copilot - 27/06/2026

For each user-selected feature/functional requirement, complete the following tasks:

Domain Testing. With the support of AI tools, apply the domain testing technique to design a comprehensive set of test cases. You are encouraged to add more test cases where needed for thorough coverage. Provide a detailed, step-by-step explanation of how you applied the technique. Review the course lectures before you begin.

Boundary Value Analysis. Likewise, apply the boundary value analysis technique to design a comprehensive set of test cases, adding more as needed for thorough coverage. Provide a detailed, step-by-step explanation of how you applied the technique. Again, review the course lectures beforehand.

Build Agent Skills that apply the domain testing and boundary value analysis techniques, so that they can be reused on additional features in future testing tasks.

2 AI Output

```
.agents/skills/bva/SKILL.md
```

```
---
name: bva
description: >
  Apply Boundary Value Analysis (BVA) to a functional requirement: identify
```

every input variable with explicit boundaries, generate OFF / ON / IN test points for each boundary, and produce test cases that probe the exact edges where defects are most likely. Reusable for any EShop feature.

applyTo: '**'

Skill: Boundary Value Analysis (BVA)

Purpose

Generate test cases that target the edges of valid input ranges. Software defects cluster at boundaries (off-by-one errors, fence-post errors, threshold miscalculations). BVA forces tests at the exact boundary, one unit below, and one unit above – guaranteeing coverage of these high-risk zones.

This skill applies ****3-value BVA**** (OFF, ON, IN per boundary), which is more thorough than the classic 2-value variant (ON + OFF only).

Methodology (Step-by-Step)

Step 1 – Identify Variables with Boundaries

Scan the requirement for every constraint that implies a numeric edge:

Signal in spec	Boundary type	Example
-----	-----	-----
` ≥ N`	closed minimum	password ≥ 8 chars → ON = 8 is **Accepted**
` > N`	open minimum	quantity > 0 → ON = 0 is **Rejected**
` ≤ N`	closed maximum	name ≤ 255 chars → ON = 255 is **Accepted**
` < N`	open maximum	age < 18 → ON = 18 is **Rejected**
"after N attempts"	threshold	lock after 3 consecutive fails → threshold = 3
"expires after N seconds/days"	time threshold	unlock after 30 s → min-wait = 30
"must be ≥ 1"	minimum count	special char count ≥ 1 → min = 1

Do ****not**** try to apply BVA to unordered/categorical data (enum values, email format flags) – use Domain Testing (EP) for those.

Step 2 – Characterise Each Boundary

For every identified boundary fill in this table:

#	Variable	Constraint	Boundary value	Step size	Effect at boundary
---	-----	-----	-----	-----	-----
B1	Password length	≥ 8 chars	8	1 char	Accepted when ≥ 8
B2	Failed attempts	triggers lock at ≥ 3	3	1 count	Locked when count = 3
B3	Lock duration	unlocks after ≥ 30 s	30	1 second	Unlock when elapsed ≥ 30 s

Step 3 – Generate Test Points per Boundary

Closed minimum boundary (accepted when value ≥ min)

Symbol	Test value	Expected behaviour
-----	-----	-----
OFF	min - 1	Rejected (below minimum)
ON	min	Accepted (exactly at minimum)
IN	min + 1	Accepted (just inside valid range)

Open minimum boundary (accepted when value > min)

Symbol	Test value	Expected behaviour
-----	-----	-----
OFF	min - 1	Rejected (below boundary)
ON	min	Rejected (boundary itself excluded)
IN	min + 1	Accepted (just inside valid range)

Closed maximum boundary (accepted when value ≤ max)

Symbol	Test value	Expected behaviour
-----	-----	-----
IN	max - 1	Accepted (just inside valid range)
ON	max	Accepted (exactly at maximum)
OFF	max + 1	Rejected (above maximum)

Open maximum boundary (accepted when value < max)

Symbol	Test value	Expected behaviour
-----	-----	-----
IN	max - 1	Accepted (just inside valid range)
ON	max	Rejected (boundary itself excluded)
OFF	max + 1	Rejected (above boundary)

Threshold boundary (behaviour changes at $\geq$ threshold)

A threshold boundary is structurally identical to a closed minimum boundary – apply the same OFF / ON / IN pattern.

Symbol	Test value	Expected behaviour
-----	-----	-----
OFF	threshold - 1	Behaviour NOT triggered
ON	threshold	Behaviour triggered
IN	threshold + 1	Behaviour still triggered

Step 4 – Create Test Cases

One test case per boundary test point:

- Set the boundary variable to the test point value.
- Keep all other variables at their valid representative (IN) values.
- Describe the exact expected outcome: accepted/rejected, action triggered/not.

For **state-based boundaries** (attempt counters, timers) the test case must describe the setup steps that bring the system to the required state (e.g., "perform 2 failed logins before this step").

Step 5 – Coverage Checklist

- [] Every boundary has OFF, ON, IN test points – for open boundaries verify ON is marked Rejected

- [] Both sides of every boundary are tested
- [] Cumulative/state boundaries describe the exact sequence to reach the state
- [] Time-based boundaries include a waiting step with precise duration
- [] Combined constraint variables (e.g., password length AND character rules) have separate BVA test cases per constraint

Output Format

1. Summary File

Create: `tests/test-cases/[feature-folder]/TC-[FEATURE]-BVA.md`

Structure (boundary analysis + summary table; ****no detailed steps****):

```markdown

# TC-[FEATURE]-BVA: Boundary Value Analysis – [Feature Name]

## Requirement ID

FR-XX

## How BVA Was Applied

[Explain which boundaries were found, boundary values, and test points generated]

## Boundary Analysis

### Boundary B1: [Variable] – [Constraint]

|                |                       |  |
|----------------|-----------------------|--|
| Attribute      | Value                 |  |
| -----          | -----                 |  |
| Variable       | ...                   |  |
| Boundary type  | min / max / threshold |  |
| Boundary value | N                     |  |
| Step size      | 1                     |  |

| Test Point | Test Value | Expected |

|  |           |  |       |  |          |  |
|--|-----------|--|-------|--|----------|--|
|  | -----     |  | ----- |  | -----    |  |
|  | OFF (N-1) |  | ...   |  | Rejected |  |
|  | ON (N)    |  | ...   |  | Accepted |  |
|  | IN (N+1)  |  | ...   |  | Accepted |  |

> \*\*Max boundary:\*\* reverse row order and swap Accepted ↔ Rejected  
> (IN = N-1 Accepted, ON = N Accepted, OFF = N+1 Rejected).

\_(repeat for each boundary)\_

## ## Test Cases Summary

|  |                  |  |             |  |          |  |       |  |          |  |
|--|------------------|--|-------------|--|----------|--|-------|--|----------|--|
|  | TC ID            |  | Description |  | Boundary |  | Point |  | Expected |  |
|  | -----            |  | -----       |  | -----    |  | ----- |  | -----    |  |
|  | TC-[FEATURE]-NNN |  | [desc]      |  | B1       |  | OFF   |  | Fail     |  |
|  | ...              |  | ...         |  | ...      |  | ...   |  | ...      |  |

## ### 2. Individual Test Case Files

Create one file per test case: `tests/test-cases/[feature-folder]/TC-[FEATURE]-NNN.md`

NNN \*\*continues from the last DT sequence number\*\* (e.g. if DT wrote `TC-LOGIN-001` ... `TC-LOGIN-010`, BVA starts from `TC-LOGIN-011`).

Structure:

```markdown

TC-[FEATURE]-NNN: [Short Description]

Requirement ID

FR-XX

Module / Test type / Technique

[Module] / Functional / BVA

Preconditions

- [list]

Test Data

| Field | Value | |
|---------|---------|--|
| ----- | ----- | |
| [field] | [value] | |

Test Steps

1. ...

Expected Result

...

Status / Related Bugs

Not Run / None

```\n

---\n

## ## Full Example: FR-02 Login – Account Lock

### ### Step 1 – Boundaries found

```\n

B1: Consecutive failed login attempts → lock at ≥ 3 (threshold = 3, step = 1)
B2: Lock duration → unlock after ≥ 30 seconds (min-wait = 30, step = 1 s)
```\n

### ### Step 2 – Test points

**\*\*B1 – Failed attempts (threshold = 3):\*\***

| Point | Attempts | Expected |  |
|-------|----------|----------|--|
| ----- | -----    | -----    |  |

|         |                          |                                   |
|---------|--------------------------|-----------------------------------|
| OFF (2) | 2 consecutive failures   | Account NOT locked                |
| ON (3)  | 3rd consecutive failure  | Account locked, 30 s timer starts |
| IN (4)  | 4th attempt while locked | Remains locked                    |

**\*\*B2 – Lock duration (min-wait = 30 s):\*\***

| Point      | Elapsed              | Expected              |
|------------|----------------------|-----------------------|
| OFF (29 s) | Wait 29 s after lock | Login still blocked   |
| ON (30 s)  | Wait 30 s after lock | Login attempt allowed |
| IN (31 s)  | Wait 31 s after lock | Login attempt allowed |

### ### Step 3 – Result

6 test cases: TC-LOGIN-NNN (continuing from DT sequence)

Summary in `tests/test-cases/login/TC-LOGIN-BVA.md`; individual files at `TC-LOGIN-NNN.md` for each test point.

.agents/skills/bva/SKILL.md

```

name: domain-testing
description: >
 Apply the Domain Testing technique to a functional requirement: identify
 every input variable's domain, partition it into valid and invalid
 equivalence classes, locate ON/OFF/IN/OUT boundary points, and generate
 a comprehensive test suite. Reusable for any EShop feature.
applyTo: '**'

```

# Skill: Domain Testing

## Purpose

Systematically design test cases for any feature by partitioning each input variable's domain into equivalence classes and probing representative values – especially at boundaries. The technique maximises fault detection while minimising redundant tests.

---

## ## Methodology (Step-by-Step)

### ### Step 1 – Identify All Input Variables

Parse the requirement (README.md / api\_specification.md) and list every input the feature processes:

- Form fields (text, number, date, select, password)
- URL / query parameters
- State variables (login status, account lock counter, attempt count)
- Implicit constraints (role, token validity)

For each variable record:

| Attribute      | Example                                             |
|----------------|-----------------------------------------------------|
| -----          | -----                                               |
| Name           | Password                                            |
| Data type      | string                                              |
| Required?      | Yes                                                 |
| Format rules   | ≥ 8 chars, ≥1 upper, ≥1 lower, ≥1 digit, ≥1 special |
| Business rules | Must match Confirm Password field                   |

---

### ### Step 2 – Define Each Variable's Domain

Write the domain formally so boundaries are unambiguous:

...

Variable: Password

Domain:

- type: string
- required: true
- length ≥ 8
- contains ≥ 1 uppercase (A-Z)
- contains ≥ 1 lowercase (a-z)
- contains ≥ 1 digit (0-9)

- contains  $\geq 1$  special (@, \$, !, %, \*, ?, &)

---

### ### Step 3 – Create Equivalence Partitions

For every variable define:

**\*\*Valid partition (EP-V):\*\*** values the system must accept.

**\*\*Invalid partitions (EP-I):\*\*** one partition per violated rule.

| Label      | Type    | Rule Violated | Representative Value           |
|------------|---------|---------------|--------------------------------|
| -----      | -----   | -----         | -----                          |
| EP-[VAR]-1 | Valid   | –             | typical valid value            |
| EP-[VAR]-2 | Invalid | [rule]        | value violating only that rule |
| ...        | ...     | ...           | ...                            |

Naming convention: `EP-[Variable initials]-[number]`

Examples: `EP-P-3` = Password partition 3, `EP-E-2` = Email partition 2.

Rules for completeness:

- One invalid partition per each distinct validation rule
- Add partitions for structural problems: empty, null, wrong type
- Add partitions for business-logic violations: duplicate, mismatch, non-existent

---

### ### Step 4 – Identify ON / OFF / IN / OUT Points

For variables with numeric or length boundaries:

| Symbol         | Position                    | Description                                           |
|----------------|-----------------------------|-------------------------------------------------------|
| -----          | -----                       | -----                                                 |
| <b>**ON**</b>  | At boundary                 | Exactly at the edge – `min` or `max` itself           |
| <b>**OFF**</b> | Just outside                | One step beyond the boundary – `min - 1` or `max + 1` |
| <b>**IN**</b>  | Well inside valid domain    | Typical valid value far from boundary                 |
| <b>**OUT**</b> | Well outside invalid domain | Obviously invalid value far from boundary             |

Treat ON and OFF as additional test cases on top of the partition-based ones.

---

### ### Step 5 – Design Test Cases (Single Fault Assumption)

For each test case, vary **exactly one** variable to an invalid partition while keeping all others at their valid IN-point values. This isolates failures to a specific input.

Ordering:

1. **TC-[FEATURE]-001**: All variables valid → expected pass (base case)
2. **TC-[FEATURE]-002 ... N**: One invalid partition per case, rest valid

---

### ### Step 6 – Coverage Checklist

Before finalising, confirm:

- [ ] At least one success case (all valid)
- [ ] Every invalid partition for every variable has a test case
- [ ] Every required field tested as empty
- [ ] All state-based conditions (locked, duplicate, expired) covered
- [ ] ON, OFF, and OUT points tested for every numeric/length boundary

---

## ## Output Format

### ### 1. Summary File

Create: `tests/test-cases/[feature-folder]/TC-[FEATURE]-DT.md`

Structure (analysis + summary table; **no detailed steps**):

```markdown

TC-[FEATURE]-DT: Domain Testing – [Feature Name]

Requirement ID

FR-XX

How Domain Testing Was Applied

[Bullet-point explanation: variables found, partitions created, test strategy]

Input Domain Analysis

Variable: [Name]

| Partition | Type | Description | Representative Value |
|-----------|---------|-----------------|----------------------|
| ----- | ----- | ----- | ----- |
| EP-X-1 | Valid | All rules met | [value] |
| EP-X-2 | Invalid | [rule violated] | [value] |

(repeat for each variable)

Test Cases Summary

| TC ID | Description | Partition Exercised | Expected |
|------------------|-------------|---------------------|----------|
| ----- | ----- | ----- | ----- |
| TC-[FEATURE]-001 | All valid | All EP-V | Pass |
| ... | ... | ... | ... |

2. Individual Test Case Files

Create one file per test case: `tests/test-cases/[feature-folder]/TC-[FEATURE]-NNN.md`

NNN is a **shared sequence** across all DT and BVA test cases for the feature (e.g. `TC-LOGIN-001`, `TC-LOGIN-002` ...). DT cases are numbered first; BVA cases follow.

Structure:

```markdown

# TC-[FEATURE]-NNN: [Short Description]

## Requirement ID

FR-XX

## Module / Test type / Technique

[Module] / Functional / Domain Testing (EP + BVA)

## Preconditions

- [list]

## Test Data

|         |         |  |
|---------|---------|--|
| Field   | Value   |  |
| -----   | -----   |  |
| [field] | [value] |  |

## Test Steps

1. ...

## Expected Result

...

## Status / Related Bugs

Not Run / None

\\`

---

## Full Example: FR-01 Register

### Step 1 – Variables

Name, Email, Password, Confirm Password

### ### Step 2 – Domains

```

Name: required string, non-empty
Email: required string, valid format (RFC 5322 subset), unique in DB
Password: required string, ≥8 chars, ≥1 upper, ≥1 lower, ≥1 digit, ≥1 special
Confirm Password: required string, must equal Password

```

### ### Step 3 – Partitions

| Label   | Variable   | Type    | Violated Rule      | Representative Value   |
|---------|------------|---------|--------------------|------------------------|
| EP-N-1  | Name       | Valid   | –                  | "Nguyen Van A"         |
| EP-N-2  | Name       | Invalid | required, empty    | " "                    |
| EP-N-3  | Name       | Invalid | non-whitespace     | " "                    |
| EP-E-1  | Email      | Valid   | –                  | "newuser@test.com"     |
| EP-E-2  | Email      | Invalid | uniqueness         | "test@eshop.com" (dup) |
| EP-E-3  | Email      | Invalid | format (no @)      | "userdomain.com"       |
| EP-E-4  | Email      | Invalid | format (no domain) | "user@"                |
| EP-E-5  | Email      | Invalid | required, empty    | " "                    |
| EP-P-1  | Password   | Valid   | –                  | "Password1!"           |
| EP-P-2  | Password   | Invalid | length < 8         | "Pa1@"                 |
| EP-P-3  | Password   | Invalid | no uppercase       | "password1!"           |
| EP-P-4  | Password   | Invalid | no lowercase       | "PASSWORD1!"           |
| EP-P-5  | Password   | Invalid | no digit           | "Password!!"           |
| EP-P-6  | Password   | Invalid | no special char    | "Password12"           |
| EP-P-7  | Password   | Invalid | required, empty    | " "                    |
| EP-CP-1 | Confirm PW | Valid   | –                  | "Password1!"           |
| EP-CP-2 | Confirm PW | Invalid | mismatch           | "Different1!"          |
| EP-CP-3 | Confirm PW | Invalid | required, empty    | " "                    |

### ### Step 4 – Boundary Points (Password length, $8 \leq \text{length} \leq 64$ )

When a variable has both a minimum and maximum boundary, test all four points for each boundary. The IN point is shared between both boundaries.

| Point           | Value     | Boundary | Expected |
|-----------------|-----------|----------|----------|
| OUT (far below) | 2 chars   | min      | Invalid  |
| OFF (below min) | 7 chars   | min      | Invalid  |
| ON (at min)     | 8 chars   | min      | Valid    |
| IN (mid-range)  | 20 chars  | —        | Valid    |
| ON (at max)     | 64 chars  | max      | Valid    |
| OFF (above max) | 65 chars  | max      | Invalid  |
| OUT (far above) | 100 chars | max      | Invalid  |

### Step 5 – Test Cases

15 test cases: TC-REGISTER-001 (all valid) + 14 covering each invalid partition.

Summary in `tests/test-cases/register/TC-REGISTER-DT.md`; individual files at `TC-REGISTER-001.md` ... `TC-REGISTER-015.md`.

3 Verdict

VALID

4 Reasoning

N/A

5 Student Fix

No changes needed — AI output is valid and complete as-is.

---

Audit-02

Anchor ID: #audit-02

# 1 Prompt + Tool

GitHub Copilot - 27/06/2026

i have updated the test-write prompt with this "- tests/test-cases/[feature]/TC-[FEATURE]-DT.md — Domain Testing test suite summary

tests/test-cases/[feature]/TC-[FEATURE]-BVA.md — BVA test suite summary

tests/test-cases/[feature]/TC-[FEATURE]-NNN.md — Detailed test suite (BVA and DT)"

help me to enhance the skill and other prompts + agents file also. you may need to ask me to confirm things that you arent sure. use context7

## 2 AI Output

.github/agents/qa.agent.md

```

description: >
 QA agent for the EShop SUT project. Designs and reviews test cases using
 structured testing techniques. Invokes domain-testing and bva skills to
 systematically cover any functional requirement.
agents: ['*']
model: Claude Sonnet 4.6 (copilot)
skills:
 - domain-testing
 - bva
 - test-writer
 - test-runner

```

# QA Agent — EShop SUT

You are the QA agent for EShop. Your role is to design comprehensive test cases for features specified in `README.md` and `api\_specification.md`.

## Workflow

1. **\*\*Read the requirement\*\*** from `README.md` (FR-XX section) and `api\_specification.md`.
2. **\*\*Apply Domain Testing\*\*** (invoke `domain-testing` skill) to identify

- all input variables, create equivalence partitions, and generate test cases.
3. **\*\*Apply BVA\*\*** (invoke `bva` skill) to identify bounded variables and generate OFF/ON/IN boundary test cases.
  4. **\*\*Write summary files\*\*** – `TC-[FEATURE]-DT.md` (analysis + table, no steps) and `TC-[FEATURE]-BVA.md` (boundary analysis + table, no steps).
  5. **\*\*Write individual test case files\*\*** – one `TC-[FEATURE]-NNN.md` per test case with full steps; NNN is a shared sequence across DT and BVA (DT first).
  6. **\*\*Update\*\*** `tests/test-summary/traceability-matrix.md` with new entries.

## ## Project Conventions

- Base URL: `http://localhost:3000`
- Default test user: `test@eshop.com` / `Test1234!`
- Default admin: `admin@eshop.com` / `Admin123!`
- Test case IDs: `TC-[FEATURE]-NNN` (shared sequence; DT cases first, BVA cases follow)
- All test cases: Status = `Not Run / None` when first created
- Language: Vietnamese labels, English IDs/values

.github/prompts/test-plan.prompt.md

---

agent: qa

---

You are the QA lead for the EShop SUT project.

## ## Task

Create (or update) a test plan document for the feature(s) specified by the user.

## ## Steps

1. **\*\*Read requirements\*\*** – scan `README.md` for every FR in scope and `api\_specification.md` for the matching endpoints.
2. **\*\*Determine scope\*\*** – list which FRs and API endpoints are in / out of scope.
3. **\*\*Identify test techniques\*\*** – for each FR note which techniques apply:
  - Domain Testing (EP) – always applicable to form/API inputs.
  - BVA – applicable when the FR defines numeric bounds, thresholds, or timers.
4. **\*\*List test case stubs\*\*** – for each technique, enumerate the test case IDs that will be written (use

`TC-[FEATURE]-DT-NNN` / `TC-[FEATURE]-BVA-NNN`).

5. **\*\*Write the plan\*\*** to `tests/test-cases/[feature]/TEST-PLAN-[FEATURE].md` using the structure below.

## ## Output Structure

```markdown

Test Plan – [Feature Name]

Scope

****In scope:**** FR-XX, FR-YY ...

****Out of scope:**** ...

Test Environment

| Item | Value |
|---------------|-----------------------------|
| ----- | ----- |
| Backend URL | http://localhost:3000 |
| Frontend URL | http://localhost:5173 |
| Admin URL | http://localhost:5174 |
| Default user | test@eshop.com / Test1234! |
| Default admin | admin@eshop.com / Admin123! |

Techniques

| Technique | Applicable FRs | Rationale |
|---------------------|----------------|--|
| ----- | ----- | ----- |
| Domain Testing (EP) | FR-XX | Input variables with equivalence classes |
| BVA | FR-XX | Numeric bounds / thresholds defined |

Test Case Inventory

> NNN is a shared sequence across DT and BVA test cases (DT cases first).

| ID | Technique | Description | Status |
|------------------|----------------|------------------|---------|
| ----- | ----- | ----- | ----- |
| TC-[FEATURE]-001 | Domain Testing | All valid inputs | Not Run |
| ... | ... | ... | ... |

Entry / Exit Criteria

****Entry:**** Backend running at localhost:3000; test accounts seeded.
****Exit:**** All planned test cases executed; defects logged.
``

Project Conventions

- Feature folder names are lowercase hyphenated (e.g. `login`, `forgot-password`, `profile`).
- Status values: `Not Run`, `Pass`, `Fail`, `Blocked`.

.github/prompts/test-run.prompt.md

agent: qa

You are the QA engineer executing a test run for the EShop SUT project.

Task

Run the test cases specified by the user and record results in the sprint test-run log.

Steps

1. ****Identify test cases to run**** – list all `TC-[FEATURE]-NNN.md` individual files under `tests/test-cases/[feature]/`. Cross-reference `TC-[FEATURE]-DT.md` and `TC-[FEATURE]-BVA.md` summaries for technique context and partition/boundary details.
2. ****Check prerequisites**** – confirm the backend is reachable at `http://localhost:3000` and the frontend at `http://localhost:5173`.
3. ****Execute each test case**** in order:
 - Follow the Preconditions, Test Data, and Test Steps exactly as written.
 - Use the Playwright browser tools to interact with the UI when the test involves a frontend step.
 - Use direct API calls (`fetch` or `curl`) for pure backend/API test cases.
4. ****Record results**** – for each test case append a row to `tests/test-runs/sprint-[N]-test-run.md`:

```
```markdown
| TC-ID | Description | Result | Notes / Bug |
```
```

Result values: `Pass`, `Fail`, `Blocked`, `Skip`.

5. ****Update the traceability matrix**** – set the `Result` and `Status` columns in `tests/test-summary/traceability-matrix.md` for every executed test case.

Test Accounts

| Role | Email | Password |
|-------|-----------------|-----------|
| ----- | ----- | ----- |
| User | test@eshop.com | Test1234! |
| Admin | admin@eshop.com | Admin123! |

Test-Run Log Format

`tests/test-runs/sprint-[N]-test-run.md`:

```markdown

# Sprint [N] Test Run – [Date]

## ## Summary

| Total | Pass  | Fail  | Blocked | Skip  |
|-------|-------|-------|---------|-------|
| ----- | ----- | ----- | -----   | ----- |
| N     | N     | N     | N       | N     |

## ## Results

| TC ID | Feature | Description | Result | Notes / Bug |
|-------|---------|-------------|--------|-------------|
| ----- | -----   | -----       | -----  | -----       |
| ...   | ...     | ...         | ...    | ...         |

## ## Notes

- If a test **\*\*Fails\*\***, describe the actual result and note it as a potential bug.
- If a test is **\*\*Blocked\*\***, explain which dependency is missing.
- Do not mark a test `Pass` unless every step and expected result is satisfied exactly.

.github/prompts/test-write.prompt.md

---

agent: qa

---

You are the QA engineer for the EShop SUT project.

## ## Task

Write a complete test suite for the feature specified by the user (e.g. "FR-02 Login").

## ## Steps

1. **\*\*Read the requirement\*\*** – open `README.md` and locate the FR section for the requested feature. Also check `api\_specification.md` for the relevant endpoints.
2. **\*\*Apply Domain Testing\*\*** – follow the `domain-testing` skill:
  - Identify all input variables and their domains.
  - Create equivalence partitions (valid + one per violated rule).
  - Include ON/OFF/IN/OUT boundary points for every numeric/length constraint.
  - Write the analysis + summary table (no detailed steps) to `tests/test-cases/[feature]/TC-[FEATURE]-DT.md`.
3. **\*\*Apply BVA\*\*** – follow the `bva` skill:
  - Identify every bounded variable (min, max, threshold, timer).
  - Generate OFF / ON / IN test points for each boundary.
  - Write the boundary analysis + summary table (no detailed steps) to `tests/test-cases/[feature]/TC-[FEATURE]-BVA.md`.
4. **\*\*Create individual test case files\*\*** – for every test case in both DT and BVA:
  - One file per test case: `tests/test-cases/[feature]/TC-[FEATURE]-NNN.md`
  - NNN is a **\*\*shared sequence\*\*** across DT and BVA (DT cases first, BVA cases follow).
  - Include: Requirement ID, Module/Test type/Technique, Preconditions, Test Data, Test Steps, Expected Result, Status.
5. **\*\*Update the traceability matrix\*\*** – add rows to `tests/test-summary/traceability-matrix.md` for every new test case.

## ## Project Conventions

- Base URL: `http://localhost:3000`
- Default test user: `test@eshop.com` / `Test1234!`

- Default admin: `admin@eshop.com` / `Admin123!`
- Test case ID format: `TC-[FEATURE]-NNN` (shared sequence across DT and BVA; DT cases first)
- Status for new cases: `Not Run / None`
- Labels in Vietnamese; IDs and values in English
- Keep all other variables at a valid IN-point when exercising one invalid partition (single fault assumption)

## ## Output Files

- `tests/test-cases/[feature]/TC-[FEATURE]-DT.md` – Domain Testing test suite summary
- `tests/test-cases/[feature]/TC-[FEATURE]-BVA.md` – BVA test suite summary
- `tests/test-cases/[feature]/TC-[FEATURE]-NNN.md` – Detailed test suite (BVA and DT)
- `tests/test-summary/traceability-matrix.md` – updated with new rows

## 3 Verdict

VALID

## 4 Reasoning

N/A

## 5 Student Fix

No changes needed — AI output is valid and complete as-is.

---

# Audit-03

Anchor ID: #audit-03

## 1 Prompt + Tool

GitHub Copilot - 27/06/2026

/test-plan I need to write test cases for FR-06, FR-09, FR-17

## 2 AI Output

TEST-PLAN-PRODUCT-DETAIL.md

# Test Plan – Product Detail (FR-06)

## Scope

**\*\*In scope:\*\*** FR-06 (Xem chi tiết sản phẩm), FR-23 (Breadcrumb), FR-24 (Image alt text), FR-21/SEC-04 (Safe display)

**\*\*Out of scope:\*\*** FR-05 (Danh sách sản phẩm), FR-07 (Giỏ hàng – kiểm tra phía Cart page)

## Test Environment

| Item          | Value                       |
|---------------|-----------------------------|
| Backend URL   | http://localhost:3000       |
| Frontend URL  | http://localhost:5173       |
| Admin URL     | http://localhost:5174       |
| Default user  | test@eshop.com / Test1234!  |
| Default admin | admin@eshop.com / Admin123! |

## Techniques

| Technique Rationale           | Applicable FRs                                                                  |
|-------------------------------|---------------------------------------------------------------------------------|
| Domain Testing (EP) partition | FR-06<br>`product_id` (URL param) và `quantity` (form input) có nhiều           |
| BVA                           | FR-06<br>`quantity` có ngưỡng tối thiểu xác định (min = 1)                      |
| GUI / Security này            | FR-23, FR-24, FR-21   Yêu cầu giao diện và bảo mật áp dụng trực tiếp trên trang |

## Input Variables

### V1 – product\_id (URL parameter)

| Domain  | Partition | Representative Value | Mô tả                        |
|---------|-----------|----------------------|------------------------------|
| Valid   | EP-ID-1   | `1`                  | ID sản phẩm tồn tại trong DB |
| Invalid | EP-ID-2   | `9999`               | ID không tồn tại trong DB    |
| Invalid | EP-ID-3   | `abc`                | ID không phải số nguyên      |

### V2 – quantity (form field, số lượng thêm vào giỏ)

| Domain  | Partition | Representative Value | Mô tả                                                         |
|---------|-----------|----------------------|---------------------------------------------------------------|
| Valid   | EP-Q-1    | `2`                  | Số nguyên dương $\geq 1$ (IN point – không phải ranh giới)    |
| Invalid | EP-Q-2    | `0`                  | Bằng 0 – cũng là BVA OFF quantity (xem TC-PRODUCT-DETAIL-005) |
| Invalid | EP-Q-3    | `-1`                 | Số âm                                                         |
| Invalid | EP-Q-4    | `1.5`                | Số thập phân                                                  |
| Invalid | EP-Q-5    | `""` (rỗng)          | Ô nhập bỏ trống                                               |
| Invalid | EP-Q-6    | `"abc"`              | Chuỗi ký tự không phải số                                     |

## ## Test Case Inventory

- > NNN là chuỗi số dùng chung cho DT và BVA (DT trước, BVA theo sau).
- > TC đánh dấu ★ phục vụ đồng thời như điểm BVA OFF; không tạo TC BVA riêng cho điểm đó.

| ID                    | Technique      | Mô tả                                                               |
|-----------------------|----------------|---------------------------------------------------------------------|
| Status                |                |                                                                     |
| TC-PRODUCT-DETAIL-001 | Domain Testing | Product ID hợp lệ → hiển thị đầy đủ: ảnh, tên, giá, mô tả, danh mục |
| TC-PRODUCT-DETAIL-002 | Domain Testing | Product ID không tồn tại (9999) → hiển thị thông báo lỗi phù hợp    |

|                                                    |                                                               |
|----------------------------------------------------|---------------------------------------------------------------|
| hợp                                                | Not Run                                                       |
| TC-PRODUCT-DETAIL-003   Domain Testing             | Product ID không phải số ("abc") → hiển thị lỗi phù           |
| hợp                                                | Not Run                                                       |
| TC-PRODUCT-DETAIL-004   Domain Testing             | Quantity hợp lệ (2) → thêm vào giỏ thành công; hiển thị toast |
| notification và/hoặc badge giỏ hàng cập nhật       | Not Run                                                       |
| TC-PRODUCT-DETAIL-005   Domain Testing             | Quantity = 0 → bị từ chối (không thêm được vào giỏ) ★BVA OFF  |
| quantity                                           | Not Run                                                       |
| TC-PRODUCT-DETAIL-006   Domain Testing             | Quantity = số âm (-1) → bị từ                                 |
| chối                                               | Not Run                                                       |
| TC-PRODUCT-DETAIL-007   Domain Testing             | Quantity = số thập phân (1.5) → bị từ chối (hệ thống chỉ nhận |
| số nguyên dương theo FR-06)                        | Not Run                                                       |
| TC-PRODUCT-DETAIL-008   Domain Testing             | Quantity = rỗng → bị từ chối khi bấm "Thêm vào giỏ            |
| hàng"                                              | Not Run                                                       |
| TC-PRODUCT-DETAIL-009   Domain Testing             | Quantity = "abc" (chuỗi ký tự không phải số) → bị từ          |
| chối                                               | Not Run                                                       |
| TC-PRODUCT-DETAIL-010   BVA                        | Quantity = 1 (ON, đúng min=1) → được chấp nhận; thêm vào giỏ  |
| thành công                                         | Not Run                                                       |
| TC-PRODUCT-DETAIL-011   BVA                        | Quantity = 2 (IN, vừa trên min) → được chấp                   |
| nhận                                               | Not Run                                                       |
| TC-PRODUCT-DETAIL-012   FR-08 / api spec           | Chưa đăng nhập, bấm "Thêm vào giỏ hàng" → bị từ chối (yêu cầu |
| đăng nhập, API trả về 401)                         | Not Run                                                       |
| TC-PRODUCT-DETAIL-013   FR-23 / GUI                | Trang `/products/:id` hiển thị breadcrumb (ví dụ: Trang chủ > |
| Chi tiết sản phẩm)                                 | Not Run                                                       |
| TC-PRODUCT-DETAIL-014   FR-24 / GUI                | Ảnh sản phẩm trên trang chi tiết có thuộc tính `alt` không    |
| rỗng                                               | Not Run                                                       |
| TC-PRODUCT-DETAIL-015   FR-21 / SEC-04             | Tên/mô tả sản phẩm chứa ký tự HTML đặc biệt (<script>, <img>) |
| được hiển thị an toàn (escaped), không render HTML | Not Run                                                       |

## ## Entry / Exit Criteria

**\*\*Entry:\*\*** Backend chạy tại localhost:3000; ít nhất 1 sản phẩm đã được seed vào DB; người dùng đã đăng nhập với tài khoản mặc định (trừ TC-PRODUCT-DETAIL-012 dùng session chưa đăng nhập).

**\*\*Exit:\*\*** Tất cả 15 test case đã được thực thi; defect (nếu có) đã được ghi nhận trong issue tracker.

## TEST-PLAN-COUPON.md

### # Test Plan – Mã Giảm Giá / Apply Coupon (FR-09)

## ## Scope

**\*\*In scope:\*\*** FR-09 (Áp dụng mã giảm giá tại bước Checkout)  
**\*\*Out of scope:\*\*** FR-17 (Admin CRUD mã giảm giá), FR-08 (luồng Checkout đầy đủ)

## ## Test Environment

| Item          | Value                       |
|---------------|-----------------------------|
| Backend URL   | http://localhost:3000       |
| Frontend URL  | http://localhost:5173       |
| Admin URL     | http://localhost:5174       |
| Default user  | test@eshop.com / Test1234!  |
| Default admin | admin@eshop.com / Admin123! |

## ## Mã giảm giá mẫu (seed data)

| Mã         | Loại    | Giá trị   | Ngưỡng tối thiểu | Hạn dùng   | Số lần/người |
|------------|---------|-----------|------------------|------------|--------------|
| `SAVE10`   | percent | 10%       | 300,000 đ        | 2099-12-31 | 1            |
| `BIGBUY`   | fixed   | 50,000 đ  | 500,000 đ        | 2099-12-31 | 1            |
| `VIP100`   | fixed   | 100,000 đ | 300,000 đ        | 2099-12-31 | 2            |
| `EXPIRED`  | percent | 20%       | 100,000 đ        | 2020-01-01 | 1            |
| `INACTIVE` | percent | 5%        | 100,000 đ        | 2099-12-31 | 1            |

> **\*\*Ghi chú seed `INACTIVE`:** Tạo bằng `POST /api/admin/coupons` (code="INACTIVE", type="percent", discount\_value=5, min\_order\_amount=100000, expired\_at="2099-12-31", max\_uses\_per\_user=1) rồi cập nhật `is\_active = 0` trực tiếp trong DB hoặc qua admin panel nếu có.

## ## Lưu ý về xác thực (Authentication Note)

Endpoint `POST /api/apply-coupon` theo api\_specification.md (5.1) truyền `user\_id` trong body JSON thay vì `Authorization: Bearer` header (khác với các endpoint có bảo mật khác). TC-COUPON-008 kiểm tra bằng cách bỏ qua `user\_id` hoặc truyền giá trị không hợp lệ. Cần đối chiếu lại cơ chế xác thực thực tế khi thực thi test.

## ## Techniques

| Technique | Applicable FRs |
|-----------|----------------|
| Rationale |                |

| Domain              | Partition | Representative Value                                                                                  | Điều kiện liên quan |
|---------------------|-----------|-------------------------------------------------------------------------------------------------------|---------------------|
| Domain Testing (EP) | FR-09     | 5 điều kiện độc lập (C1-C5) × 2 loại coupon (percent/fixed) → nhiều partition                         |                     |
| BVA                 | FR-09     | `total_amount` vs `min_order_amount` có ngưỡng rõ ràng; `usage_count` có giới hạn `max_uses_per_user` |                     |

## ## Input Variables

### ### V1 – code (mã giảm giá nhập vào)

| Domain  | Partition | Representative Value | Điều kiện liên quan                      |
|---------|-----------|----------------------|------------------------------------------|
| Valid   | EP-CODE-1 | `SAVE10`             | C1: tồn tại, is_active=1                 |
| Invalid | EP-CODE-2 | `NOTEXIST`           | C1: không tồn tại trong DB               |
| Invalid | EP-CODE-3 | `INACTIVE`           | C1: is_active=0 (xem ghi chú seed data)  |
| Invalid | EP-CODE-4 | `EXPIRED`            | C2: đã hết hạn (expired_at = 2020-01-01) |

### ### V2 – total\_amount (tổng đơn hàng)

| Domain  | Partition  | Representative Value | Điều kiện liên quan                                                 |
|---------|------------|----------------------|---------------------------------------------------------------------|
| Valid   | EP-TOTAL-1 | `400,000`            | C3: trên ngưỡng SAVE10 (IN point, không phải ranh giới)             |
| Valid   | EP-TOTAL-2 | `500,000`            | C3: đúng ngưỡng BIGBUY – cũng là BVA ON BIGBUY (xem TC-COUPON-002)  |
| Invalid | EP-TOTAL-3 | `299,999`            | C3: dưới ngưỡng SAVE10 – cũng là BVA OFF SAVE10 (xem TC-COUPON-006) |
| Invalid | EP-TOTAL-4 | `0`                  | C3: bằng 0                                                          |

### ### V3 – trạng thái xác thực (C4)

| Domain  | Partition | Mô tả                              |
|---------|-----------|------------------------------------|
| Valid   | EP-AUTH-1 | Có JWT Token hợp lệ (đã đăng nhập) |
| Invalid | EP-AUTH-2 | Không có token (chưa đăng nhập)    |

### ### V4 – số lần đã dùng mã (C5)

| Domain  | Partition  | Representative Value  | Mô tả                                                                      |
|---------|------------|-----------------------|----------------------------------------------------------------------------|
| Valid   | EP-USAGE-1 | 0 (chưa dùng lần nào) | C5: dưới giới hạn max_uses_per_user=1                                      |
| Invalid | EP-USAGE-2 | 1 (đã dùng đủ lượt)   | C5: đã đạt max_uses_per_user=1 – cũng là BVA OFF usage (xem TC-COUPON-009) |

### ### V5 – loại mã giảm giá (discount\_type)

| Domain | Partition | Representative Value | Công thức                        |
|--------|-----------|----------------------|----------------------------------|
| Valid  | EP-TYPE-1 | `percent`            | discount = total × value / 100   |
| Valid  | EP-TYPE-2 | `fixed`              | discount = discount_value (flat) |

## ## Test Case Inventory

- > NNN là chuỗi số dùng chung cho DT và BVA (DT trước, BVA theo sau).
- > TC đánh dấu ★ phục vụ đồng thời như điểm BVA OFF; không tạo TC BVA riêng cho điểm đó.

| ID            | Technique      | Mô tả                                                                                                            |
|---------------|----------------|------------------------------------------------------------------------------------------------------------------|
| TC-COUPON-001 | Domain Testing | Tất cả điều kiện hợp lệ, SAVE10 (percent), total=400,000 → discount=40,000, final=360,000   Not Run              |
| TC-COUPON-002 | Domain Testing | Tất cả điều kiện hợp lệ, BIGBUY (fixed), total=500,000 → discount=50,000, final=450,000 ★BVA ON BIGBUY   Not Run |
| TC-COUPON-003 | Domain Testing | Mã không tồn tại trong DB → trả về lỗi phù hợp   Not Run                                                         |
| TC-COUPON-004 | Domain Testing | Mã bị vô hiệu hóa (INACTIVE, is_active=0) → trả về lỗi   Not Run                                                 |
| TC-COUPON-005 | Domain Testing | Mã đã hết hạn (EXPIRED, expired_at=2020-01-01) → trả về lỗi                                                      |

| TC-ID         | Tester         | Test Case Description                                                                                  | Expected Result | Actual Result | Status   |
|---------------|----------------|--------------------------------------------------------------------------------------------------------|-----------------|---------------|----------|
| TC-COUPON-006 | Domain Testing | SAVE10, total_amount=299,999 < min_order_amount (300,000) → bị từ chối                                 | Not Run         |               | lỗi      |
| ★BVA OFF      | SAVE10         |                                                                                                        | Not Run         |               |          |
| TC-COUPON-007 | Domain Testing | SAVE10, total_amount=0 → bị từ chối                                                                    | Not Run         |               |          |
| TC-COUPON-008 | Domain Testing | Chưa đăng nhập (không truyền user_id / JWT không hợp lệ) → trả về lỗi xác thực                         | Not Run         |               |          |
| TC-COUPON-009 | Domain Testing | SAVE10 dùng lần thứ 2 (usage=1=max_uses_per_user) → bị từ chối                                         | Not Run         |               | ★BVA OFF |
| TC-COUPON-010 | Domain Testing | VIP100 lần dùng thứ 2 (max_uses=2, usage=1) → vẫn được chấp nhận                                       | Not Run         |               |          |
| TC-COUPON-011 | Domain Testing | VIP100 lần dùng thứ 3 (đã đạt max_uses=2) → bị từ chối                                                 | Not Run         |               |          |
| TC-COUPON-012 | BVA            | SAVE10, total=300,000 (ON, đúng ngưỡng min=300,000) → được chấp nhận                                   | Not Run         |               |          |
| TC-COUPON-013 | BVA            | SAVE10, total=300,001 (IN, vừa trên ngưỡng SAVE10) → được chấp nhận                                    | Not Run         |               |          |
| TC-COUPON-014 | BVA            | BIGBUY, total=499,999 (OFF, dưới ngưỡng BIGBUY min=500,000) → bị từ chối                               | Not Run         |               |          |
| TC-COUPON-015 | BVA            | BIGBUY, total=500,001 (IN, vừa trên ngưỡng BIGBUY) → được chấp nhận                                    | Not Run         |               |          |
| TC-COUPON-016 | GUI            | Nhất quán đơn vị tiền: hiển thị số tiền giảm và tổng tiền có ký hiệu đ và phân cách hàng nghìn (FR-21) | Not Run         |               |          |
| TC-COUPON-017 | GUI            | Nhất quán ngôn ngữ: thông báo thành công/lỗi hiển thị bằng tiếng Việt rõ ràng (FR-21)                  | Not Run         |               |          |
| TC-COUPON-018 | Security / GUI | Bảo mật hiển thị: mã giảm giá nhập vào được hiển thị an toàn, không render HTML (SEC-04 / FR-21)       | Not Run         |               |          |

## ## Entry / Exit Criteria

**\*\*Entry:\*\*** Backend chạy tại localhost:3000; DB đã seed các mã giảm giá mẫu (SAVE10, BIGBUY, VIP100, EXPIRED); mã `INACTIVE` đã được tạo và cập nhật `is\_active=0` theo hướng dẫn seed data; tài khoản test đã tồn tại và chưa dùng bất kỳ mã nào.

**\*\*Exit:\*\*** Tất cả 15 test case đã được thực thi; defect (nếu có) đã được ghi nhận trong issue tracker.

TEST-PLAN-COUPON-ADMIN.md

# Test Plan – Quản lý Mã Giảm Giá Admin (FR-17)

## ## Scope

**\*\*In scope:\*\*** FR-17 (Admin Thêm / Xem / Xóa mã giảm giá), FR-12 (kiểm soát truy cập – áp dụng cho các endpoint `/api/admin/coupons`)

**\*\*Out of scope:\*\*** FR-09 (người dùng áp dụng mã giảm giá tại Checkout)

## ## Test Environment

| Item          | Value                       |
|---------------|-----------------------------|
| Backend URL   | http://localhost:3000       |
| Frontend URL  | http://localhost:5173       |
| Admin URL     | http://localhost:5174       |
| Default user  | test@eshop.com / Test1234!  |
| Default admin | admin@eshop.com / Admin123! |

## ## Techniques

| Technique           | Applicable FRs |                                                                                             |
|---------------------|----------------|---------------------------------------------------------------------------------------------|
| Rationale           |                |                                                                                             |
| Domain Testing (EP) | FR-17, FR-12   | Mỗi field nhập liệu đều có quy tắc hợp lệ/không hợp lệ; role và auth là partition bắt buộc  |
| BVA                 | FR-17          | `discount_value` (>0), `min_order_amount` (≥0) và `max_uses_per_user` (≥1) có ngưỡng số học |

## ## Input Variables

### ### V1 – code (mã giảm giá)

| Domain  | Partition | Representative Value | Mô tả                        |
|---------|-----------|----------------------|------------------------------|
| Valid   | EP-CODE-1 | `"SUMMER25"``        | Chuỗi duy nhất, không rỗng   |
| Invalid | EP-CODE-2 | `""`                 | Rỗng                         |
| Invalid | EP-CODE-3 | `"SAVE10"`` (đã có)  | Trùng mã đã tồn tại trong DB |

### ### V2 – type (loại giảm giá)

| Domain  | Partition | Representative Value | Mô tả                              |
|---------|-----------|----------------------|------------------------------------|
| Valid   | EP-TYPE-1 | `"percent"`          | Giảm theo phần trăm                |
| Valid   | EP-TYPE-2 | `"fixed"`            | Giảm theo số tiền cố định          |
| Invalid | EP-TYPE-3 | `"voucher"`          | Giá trị ngoài tập {percent, fixed} |

### ### V3 – discount\_value (giá trị giảm)

| Domain  | Partition | Representative Value | Mô tả                                                             |
|---------|-----------|----------------------|-------------------------------------------------------------------|
| Valid   | EP-DV-1   | `15`                 | Số dương                                                          |
| Invalid | EP-DV-2   | `0`                  | Bằng 0 – cũng là BVA OFF discount_value (xem TC-COUPON-ADMIN-006) |
| Invalid | EP-DV-3   | `-10`                | Số âm                                                             |

### ### V4 – expired\_at (ngày hết hạn)

| Domain  | Partition | Representative Value | Mô tả                       |
|---------|-----------|----------------------|-----------------------------|
| Valid   | EP-EXP-1  | `"2099-12-31"`       | Ngày tương lai hợp lệ       |
| Invalid | EP-EXP-2  | `""` (rỗng)          | Thiếu trường bắt buộc       |
| Invalid | EP-EXP-3  | `"31-12-2099"`       | Định dạng ngày không hợp lệ |

### ### V5 – min\_order\_amount (ngưỡng đơn hàng tối thiểu)

| Domain  | Partition | Representative Value | Mô tả                                                              |
|---------|-----------|----------------------|--------------------------------------------------------------------|
| Valid   | EP-MOA-1  | `0`                  | Bằng 0 (hợp lệ) – cũng là BVA ON                                   |
| Valid   | EP-MOA-2  | `200000`             | Số dương                                                           |
| Invalid | EP-MOA-3  | `-1`                 | Số âm – cũng là BVA OFF min_order_amount (xem TC-COUPON-ADMIN-010) |

### V6 – max\_uses\_per\_user (số lần dùng tối đa mỗi người)

| Domain            | Partition | Representative Value | Mô tả                                                                |
|-------------------|-----------|----------------------|----------------------------------------------------------------------|
| -----             | -----     | -----                |                                                                      |
| Valid             | EP-MPU-1  | `1`                  | Đúng ngưỡng tối thiểu (min=1) – cũng là BVA ON                       |
| max_uses_per_user |           |                      |                                                                      |
| Valid             | EP-MPU-2  | `5`                  | Số nguyên dương >                                                    |
| 1                 |           |                      |                                                                      |
| Invalid           | EP-MPU-3  | `0`                  | Bằng 0 – cũng là BVA OFF max_uses_per_user (xem TC-COUPON-ADMIN-011) |
| Invalid           | EP-MPU-4  | `-1`                 | Số âm                                                                |

### V7 – trạng thái xác thực / phân quyền

| Domain  | Partition | Mô tả                                    |
|---------|-----------|------------------------------------------|
| -----   | -----     | -----                                    |
| Valid   | EP-ROLE-1 | JWT hợp lệ, role = admin                 |
| Invalid | EP-ROLE-2 | Không có JWT (chưa đăng nhập)            |
| Invalid | EP-ROLE-3 | JWT hợp lệ nhưng role = user (non-admin) |

## Test Case Inventory

- > NNN là chuỗi số dùng chung cho DT và BVA (DT trước, BVA theo sau).
- > TC đánh dấu ★ phục vụ đồng thời như điểm BVA OFF; không tạo TC BVA riêng cho điểm đó.

| ID                  | Technique      | Mô tả                                                                      |
|---------------------|----------------|----------------------------------------------------------------------------|
| -----               | -----          | -----                                                                      |
| TC-COUPON-ADMIN-001 | Domain Testing | Tất cả trường hợp lệ (percent type) → coupon được tạo thành công   Not Run |
| TC-COUPON-ADMIN-002 | Domain Testing | Tất cả trường hợp lệ (fixed type) → coupon được tạo thành công   Not Run   |

|                     |                |                                                                                           |         |
|---------------------|----------------|-------------------------------------------------------------------------------------------|---------|
| TC-COUPON-ADMIN-003 | Domain Testing | code = rỗng → bị từ chối, thông báo lỗi rõ ràng                                           | Not Run |
| TC-COUPON-ADMIN-004 | Domain Testing | code trùng với mã đã tồn tại → bị từ chối (unique constraint)                             | Not Run |
| TC-COUPON-ADMIN-005 | Domain Testing | type = giá trị không hợp lệ ("voucher") → bị từ chối                                      | Not Run |
| TC-COUPON-ADMIN-006 | Domain Testing | discount_value = 0 → bị từ chối ★BVA OFF                                                  | Not Run |
| TC-COUPON-ADMIN-007 | Domain Testing | discount_value = số âm → bị từ chối                                                       | Not Run |
| TC-COUPON-ADMIN-008 | Domain Testing | expired_at = rỗng → bị từ chối                                                            | Not Run |
| TC-COUPON-ADMIN-009 | Domain Testing | expired_at = định dạng ngày không hợp lệ ("31-12-2099") → bị từ chối                      | Not Run |
| TC-COUPON-ADMIN-010 | Domain Testing | min_order_amount = số âm (-1) → bị từ chối ★BVA OFF                                       | Not Run |
| TC-COUPON-ADMIN-011 | Domain Testing | max_uses_per_user = 0 → bị từ chối ★BVA OFF                                               | Not Run |
| TC-COUPON-ADMIN-012 | Domain Testing | max_uses_per_user = số âm → bị từ chối                                                    | Not Run |
| TC-COUPON-ADMIN-013 | Domain Testing | Xóa coupon theo ID hợp lệ → xóa thành công                                                | Not Run |
| TC-COUPON-ADMIN-014 | Domain Testing | Xóa coupon theo ID không tồn tại → trả về lỗi (status 4xx)                                | Not Run |
| TC-COUPON-ADMIN-015 | Domain Testing | Admin lấy danh sách coupon (`GET /api/coupons`) → trả về đầy đủ các trường (kể cả mã mẫu) | Not Run |
| TC-COUPON-ADMIN-016 | Domain Testing | Chưa đăng nhập (không có JWT) → 401 Unauthorized                                          | Not Run |
| TC-COUPON-ADMIN-017 | Domain Testing | Đăng nhập với role = user (non-admin) → 403 Forbidden                                     | Not Run |
| TC-COUPON-ADMIN-018 | BVA nhận       | discount_value = 1 (ON, đúng min > 0) → được chấp                                         | Not Run |
| TC-COUPON-ADMIN-019 | BVA nhận       | discount_value = 2 (IN, vừa trên min) → được chấp                                         | Not Run |
| TC-COUPON-ADMIN-020 | BVA nhận       | min_order_amount = 0 (ON, đúng min ≥ 0) → được chấp                                       | Not Run |
| TC-COUPON-ADMIN-021 | BVA nhận       | min_order_amount = 1 (IN, vừa trên min) → được chấp                                       | Not Run |
| TC-COUPON-ADMIN-022 | BVA            | max_uses_per_user = 1 (ON, đúng min ≥ 1) → được chấp                                      |         |

```
nhận | Not Run |
| TC-COUPON-ADMIN-023 | BVA | max_uses_per_user = 2 (IN, vừa trên min) → được chấp
nhận | Not Run |
| TC-COUPON-ADMIN-024 | Domain Testing | expired_at = ngày trong quá khứ (vd: "2020-01-01") → hành vi cần
xác minh (thăm dò: chấp nhận hay từ chối?) | Not Run |
| TC-COUPON-ADMIN-025 | Domain Testing | Thiếu hoàn toàn trường bắt buộc trong body (field key không có
mặt, không phải rỗng) → bị từ chối | Not Run |
| TC-COUPON-ADMIN-026 | Domain Testing | Xử lý an toàn chuỗi nhập vào khi hiển thị mã coupon (chống
XSS) | Not Run |
```

## ## Entry / Exit Criteria

**\*\*Entry:\*\*** Backend chạy tại localhost:3000; tài khoản admin đã sẵn sàng; ít nhất 1 mã giảm giá mẫu (SAVE10) đã được seed.

**\*\*Exit:\*\*** Tất cả 26 test case đã được thực thi; defect (nếu có) đã được ghi nhận trong issue tracker.

## 3 Verdict

INCOMPLETE

## 4 Reasoning

Missing and duplicate test cases in the AI output.

## 5 Student Fix

Fix in the next prompt

---

# Audit-04

**Anchor ID:** #audit-04

## 1 Prompt + Tool

GitHub Copilot - 27/06/2026

validate the test-plan-\* (coupon, coupon-admin, product-detail) so that it strictly follow the README.md and api-spec.

## 2 AI Output

Validation Report – EShop Test Plans

### 1. TEST-PLAN-COUPON.md – NEEDS REVISION

Critical/Major issues:

| # | Severity | Issue                                                                                                                                                                        |
|---|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Major    | TC-COUPON-011 violates single-fault assumption – tests expired + below-threshold simultaneously; split into two or remove (TC-005 and TC-006 already cover these separately) |
| 2 | Major    | TC-COUPON-004 (is_active=0) has no seed data – none of the 4 seeded coupons have is_active=0; add to seed or add a setup step                                                |
| 3 | Major    | TC-COUPON-006 uses total=200,000 but EP-TOTAL-3 representative is 299,999 – misalignment breaks traceability                                                                 |
| 4 | Major    | EP-TOTAL-1 representative = 300,000 is the ON boundary, not an interior IN point – conflicts with TC-COUPON-015 BVA coverage; change to 500,000                              |
| 5 | Major    | TC-COUPON-013 is a duplicate of TC-COUPON-001 with no new partition; remove it                                                                                               |
| 6 | Minor    | TC-COUPON-017 (BVA usage=0) duplicates TC-COUPON-001 precondition                                                                                                            |
| 7 | Minor    | Auth mechanism (Authorization header vs. user_id body) is ambiguous across all TCs                                                                                           |

Missing coverage: BVA for BIGBUY threshold (500,000), test for empty string "" code, VIP100 max\_uses boundary.

### 2. TEST-PLAN-COUPON-ADMIN.md – NEEDS REVISION

Critical/Major issues:

| # | Severity | Issue                                                                                                                                                                            |
|---|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Major    | 3 duplicate pairs: TC-018=TC-006, TC-021=TC-010, TC-024=TC-011 – BVA OFF points are identical to DT invalid representatives; remove the BVA duplicates and annotate the DT cases |
| 2 | Major    | TC-COUPON-ADMIN-015 – listing endpoint is GET /api/coupons (api spec 5.2), not /api/admin/coupons; wrong URL tested                                                              |
| 3 | Minor    | TC-COUPON-ADMIN-014 asserts HTTP 404 for non-existent ID delete, but api spec does not specify this status code                                                                  |

Missing coverage: Create coupon with past expired\_at, omitted required fields (vs. empty string), non-integer max\_uses\_per\_user, full response body field verification.

### 3. TEST-PLAN-PRODUCT-DETAIL.md – NEEDS REVISION

Critical/Major issues:

| # | Severity | Issue                                                                                                                                |
|---|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Major    | EP-Q-1 defined as > 1 – but FR-06 says min is 1; quantity=1 falls outside the valid partition definition; redefine as >= 1           |
| 2 | Major    | EP-Q-6 ("abc") has no test case – declared in the partition table but no TC exercises it                                             |
| 3 | Major    | TC-PRODUCT-DETAIL-007 expected result says "rejected OR rounded" – FR-06 is unambiguous (integers only); remove the "rounded" option |
| 4 | Minor    | TC-PRODUCT-DETAIL-009 (BVA OFF, qty=0) duplicates TC-PRODUCT-DETAIL-005 (DT, qty=0)                                                  |

Missing coverage: Breadcrumb (FR-23), image alt text (FR-24), add-to-cart unauthenticated, XSS safe display (FR-21/SEC-04), explicit toast/badge assertion after add-to-cart.

### Cross-Cutting Issues (all plans)

Individual TC files are empty/missing – TC-COUPON-001.md is a blank template; no individual TC file has content yet.

No traceability matrix – tests/test-summary/traceability-matrix.md does not exist.

### Consolidated Fix Priority

| Priority | Plan           | Action                                                                  |
|----------|----------------|-------------------------------------------------------------------------|
| 1        | COUPON         | Remove TC-011; add seed data / setup for is_active=0                    |
| 2        | COUPON         | Fix TC-006 data to 299,999; change EP-TOTAL-1 representative to 500,000 |
| 3        | COUPON-ADMIN   | Merge the 3 duplicate BVA/DT pairs                                      |
| 4        | COUPON-ADMIN   | Correct listing endpoint to GET /api/coupons in TC-015                  |
| 5        | PRODUCT-DETAIL | Redefine EP-Q-1 as >= 1; add TC for EP-Q-6                              |
| 6        | PRODUCT-DETAIL | Fix TC-007 expected result to "rejected" only                           |
| 7        | ALL            | Write individual TC-NNN.md files                                        |
| 8        | ALL            | Create traceability matrix                                              |

## 3 Verdict

VALID

## 4 Reasoning

N/A

## 5 Student Fix

No changes needed – AI output is valid and complete as-is.

---

# Audit-05

Anchor ID: #audit-05

## 1 Prompt + Tool

Antigravity - 27/06/2026

[qa.agent.md] [test-plan.prompt.md] [api\_specification.md] [README.md] create test plan for 'cart' feature on mobile (FR-20)

## 2 AI Output

The output is committed to git.

9fe8a73 test: add test cases for FR-20 (cart)

## 3 Verdict

VALID

## 4 Reasoning

N/A

## 5 Student Fix

No changes needed — AI output is valid and complete as-is.

---

# Audit-06

Anchor ID: #audit-06

## 1 Prompt + Tool

Antigravity - 28/06/2026

[TEST-PLAN-PRODUCT-DETAIL.md] [qa.agent.md] write all test cases for FR-06, then update the [traceability-matrix.md]

## 2 AI Output

The output is committed to git.

37cdbc test: add test cases for FR-06

## 3 Verdict

VALID

## 4 Reasoning

N/A

## 5 Student Fix

No changes needed — AI output is valid and complete as-is.

---

# Audit-07

Anchor ID: #audit-07

## 1 Prompt + Tool

Antigravity - 28/06/2026

[TEST-PLAN-COUPON.md] [qa.agent.md] write all test cases for FR-09, then update the [traceability-matrix.md]

## 2 AI Output

The output is committed to git.

718ebef test: add test cases for FR-09

## 3 Verdict

VALID

## 4 Reasoning

N/A

## 5 Student Fix

No changes needed — AI output is valid and complete as-is.

---

# Audit-08

Anchor ID: #audit-08

## 1 Prompt + Tool

Antigravity - 28/06/2026

[TEST-PLAN-COUPON-ADMIN.md] [qa.agent.md] write all test cases for FR-17, then update the [traceability-matrix.md]

2 AI Output

The output is committed to git.

```
f359171 test: add test cases for FR-17
```

3 Verdict

VALID

4 Reasoning

N/A

5 Student Fix

No changes needed — AI output is valid and complete as-is.

Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

| Metric                               | Count | Percentage |
|--------------------------------------|-------|------------|
| Total AI-generated artifacts audited | 8     | -          |
| VALID (correct, accepted as-is)      | 7     | 87.5%      |
| INVALID (wrong; rejected)            | 0     | 0%         |
| INCOMPLETE (acceptable after edits)  | 1     | 12.5%      |

## Conclusion

AI did the work well, but there are some minor issues in the test plans that need to be addressed. The majority of the test cases are valid and complete, but a few require revisions to align with the specifications and ensure proper coverage.

## Mandatory Disclosure

“Test cases, report artifacts” was initially generated by GitHub Copilot; I reviewed and modified all submission-needed sections, added changes to files; “audit-report, privacy-checklist, self-assessment” was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

## Signature

|                 |                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| Student name :  | NGÔ NGUYỄN THÊ KHOA                                                                                          |
| Student ID:     | 23127065                                                                                                     |
| Class / Cohort: | 23KTPM3                                                                                                      |
| Course:         | CS423 / CSC13003 – Software Testing                                                                          |
| Instructor:     | Dr. Lam Quang Vu / Dr. Tran Duy Hoang / MSc. Tran Thi Bich Hanh / MSc. Truong Phuoc Loc / MSc. Ho Tuan Thanh |
| Date:           | 29/06/2026                                                                                                   |
| Signature:      | khoa                                                                                                         |

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